ECO 362 Financial Investments
Midterm Exam

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Instructions

1. Check that you have all 5 pages. There are 4 questions.

Question	Points
1	25
2	20
3	30
4	25
Total	100

2. This is a closed book exam. However, you may use a one-page cheat sheet containing formulas or anything else you may find helpful. The cheat sheet must be letter size (8.5x11in) with writing on one side only (i.e., the back side must be left blank). You are under the Honor Code to comply with this requirement.

3. You may use a writing instrument, a cheat sheet, and a calculator. Put everything else away. The calculator cannot be a phone, a tablet, or a laptop. If you do not have a calculator, you may borrow one. Please return it after the exam.

4. You may not communicate with other students or violate the Honor Code during this exam. Write your name and sign the Honor Code on the first page of the exam book:

   I pledge my honor that I have not violated the Honor Code during this examination.

5. Write your answers in the exam books. *Show all steps necessary in deriving your solution for full credit.* Report the yield, the coupon rate, or the return in percentage points. Round all answers to two decimal places (e.g., $95.67 or 1.23%).

6. You have 90 minutes to complete the exam. After 90 minutes, you must stop working on the exam. You then have an extra 20 minutes to submit your answers as one scanned pdf file or photographed images through Gradescope. If you do not have a device to scan or photograph, give your exam books to a preceptor, who will be able to do it for you.

7. In addition to submission by Gradescope, submit the hard copies of the exam books before you leave the room. We need them in case there are any technical issues (e.g., unreadable scans). Make sure that your name is on the exam books that you submit.

1. The following table reports annually compounded par yields in percentage points. For example, the annually compounded yield on a 2-year par bond is 1.61% in 2019. Assume no arbitrage.

	Remaining maturity (years)		
Year	1	2	3
2019	1.64	1.61	1.64
2020	0.12	0.12	0.18
2021	0.42	0.76	0.99

[25 points total, 5 points each part]

(a) What is the price of a 1-year zero coupon bond with a face value of $100 in 2020?

(b) What is the price of a 2-year zero coupon bond with a face value of $100 in 2020?

(c) What is your return if you buy the 3-year par bond with a face value of $100 in 2019 and sell it in 2020? Hint: Do not forget that you receive a coupon in year 2020.

(d) In 2021, what is the price of a 3-year annuity that pays $1 per year (i.e., first payment of $1 in 2022 and the last payment of $1 in 2024).

(e) You win the lottery in 2021. You can claim the prize as a one-time payment of $1 million or a 3-year annuity that pays $350,000 per year. To maximize your wealth, should you take the prize as a one-time payment or an annuity? State your reasoning clearly for full credit.

2. The annually compounded riskless interest rate is 1%. There are two possible states a year from now: expansion and recession. The probability of each state is 0.5. The following table contains the possible prices of Disney and Verizon a year from now. Assume no arbitrage.

[20 points total, 5 points each part]

		Possible prices	
Stock	Price	Expansion	Recession
Disney	\$101	\$145	\$88
Verizon		\$46	\$34

(a) The current price of Disney is \$101. What is the current price of Verizon that is consistent with no arbitrage?

(b) What is the price of a security that pays \$1 in expansion and \$0 in recession.

(c) What is the price of a security that pays \$0 in expansion and \$1 in recession.

(d) Write down one equation that shows how your answers to parts (a), (b), and (c) are related to each other by no arbitrage.

3. Each column in the following table represents a model portfolio, depending on your level of risk tolerance. Fill in the blanks so that each column adds up to 100%, and all three portfolios are consistent with the portfolio separation theorem. Assume that T-bills are riskless, and the other assets are risky. Round your answers to the nearest 1% (e.g., write 50% instead of 49.5%). If there is not enough information provided to fill in a blank, you may simply write "unknown" in the blank for full credit.

Asset	Conservative	Moderate	Aggressive
US stocks			35%
UK stocks		24%	
Japanese stocks			15%
German stocks	10%		
T-bills	50%		0

[30 points total, 3.33 points each blank]

4. Let $R_{m,t}$ be the market return at time t. The market return has a mean $\mathbb{E}[R_{m,t}] = 0.94\%$ and a standard deviation $\sigma(R_{m,t}) = 5.3\%$. The riskless interest rate is constant and equal to $R_b = 0.08\%$.

The return on stock i (either Amazon and BP) at time t follows the statistical model:

$$R_{i,t} - R_b = \beta_i(R_{m,t} - R_b) + \epsilon_{i,t}.$$

β_i is the market beta of stock i, and $\epsilon_{i,t}$ is the idiosyncratic component of returns with mean zero and standard deviation $\sigma(\epsilon_{i,t})$. Assume that $\epsilon_{i,t}$ is uncorrelated across stocks and also uncorrelated with the market return $R_{m,t}$. Thus, the variance of returns on stock i is

$$\text{Var}(R_{i,t}) = \beta_i^2\sigma(R_{m,t})^2 + \sigma(\epsilon_{i,t})^2.$$

The following table contains the market beta and the standard deviation of the idiosyncratic component of returns.

Stock	β_i	$\sigma(\epsilon_{i,t})$
Amazon	0.87	7.5%
BP	1.53	4.5%

[25 points total, 5 points each part]

(a) What is the expected excess return (i.e., $\mathbb{E}[R_{i,t} - R_b]$) on Amazon? What is the expected excess return on BP?

(b) What are the mean and the standard deviation of returns on a portfolio that invests 50% in each of Amazon and BP?

(c) What are the mean and the standard deviation of returns on a portfolio that invests 30% in Amazon and 70% in BP?

(d) Which of the two portfolios in parts (b) and (c) cannot be the tangency portfolio? State your reasoning clearly for full credit.

(e) Is the minimum variance portfolio equal to a portfolio that invests 100% in Amazon and 0% in BP? State your reasoning clearly for full credit.